

IDENTIFYING INFORMATION:

NAME: Arienti, Marco

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0001-8166-0016>

POSITION TITLE: Principal Member of Technical Staff

PRIMARY ORGANIZATION AND LOCATION: Sandia National Laboratories, Livermore, California, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
California Institute of Technology, Pasadena, California, United States	PHD	09/1997 - 06/2003	Aeronautics
Politecnico di Milano, Milano, Not Applicable, N/A, Italy	BME	07/1989 - 09/1994	Aerospace

Appointments and Positions

2010 - present Principal Member of Technical Staff, Sandia National Laboratories, Livermore, California, United States

2004 - 2010 Technical Staff, United Technologies Corporation , East Hartford, Connecticut, United States

1997 - 2003 Research Graduate, California Institute of Technology, Pasadena, California, United States

1996 - 1997 Intern, Joint Research Center, Ispra, Not Applicable, N/A, Italy

1995 - 1996 Graduate Fellowship, European Space Agency (ESOC), Darmstadt, Not Applicable, N/A, Germany

Products

1. Arienti M, Wenzel EA., Guildenbecher DR., Beresh SJ.. Simulation and analysis of droplet aerobreakup in a ballistic wave system. AIAA Journal. 2025; 63(8):3309.
2. Arienti M, Sussman M. A numerical study of the thermal transient in high-pressure Diesel injection. International Journal of Multiphase Flow. 2017; 88:205. Available from: <https://doi.org/10.1016/j.ijmultiphaseflow.2016.09.017>
3. Wenzel EA., Arienti M. A conservative, interface-resolved framework for the modeling and simulation of compressible phase change. J. Computational Phys.. 2023; 477:111957.
4. Jemison M, Sussman M, Arienti M. Compressible, multi-phase semi-implicit method with moment of fluid interface representation. J. of Comput. Phys.. 2014; 279:182.
5. Arienti M, Pan W, Li X, Karniadakis G. Many-body dissipative particle dynamics simulation of liquid/vapor and liquid/solid interactions. J Chem Phys. 2011 May 28;134(20):204114. PubMed PMID: [21639431](#).
6. Amat MA, Arienti M, Fonoberov VA, Kevrekidis IG, Maroudas D. Coarse molecular-dynamics analysis of an order-to-disorder transformation of a krypton monolayer on graphite. J Chem

Phys. 2008 Nov 14;129(18):184106. PubMed PMID: [19045385](#).

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Biographical Sketch Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

Certified by Arienti, Marco in SciENcv on 2026-04-13 13:12:08